

Problem

Calculate the inverse Laplace transform of:

(a) $\frac{6(s-1)}{s^4-1}$

(b) $\frac{se^{-\pi s}}{s^2+1}$

(c) $\frac{8}{s(s+1)^3}$

Step-by-step solution

Step 1 of 6

(a)

The function $X(s)$ is,

$$X(s) = \frac{6(s-1)}{(s^4-1)}$$

Simplify the function $X(s)$.

$$\begin{aligned} X(s) &= \frac{6(s-1)}{(s^2+1)(s^2-1)} \\ &= \frac{6}{(s^2+1)(s+1)} \end{aligned}$$

Use partial fractions to simplify the function $X(s)$.

$$\frac{6}{(s^2+1)(s+1)} = \frac{As+B}{s^2+1} + \frac{C}{s+1} \dots\dots (1)$$

$$6 = (As+B)(s+1) + C(s^2+1)$$

$$6 = (A+C)s^2 + (A+B)s + (B+C) \dots\dots (2)$$

Step 2 of 6

Compare the constant terms in the equation (2).

$$B + C = 6 \dots\dots (3)$$

Compare the coefficients of s terms in the equation (2).

$$A + B = 0$$

$$A = -B$$

Compare the coefficients of s^2 terms in the equation (2).

$$A + C = 0$$

$$C = -A$$

$$= -(-B)$$

$$= B$$

Substitute B for C in equation (3).

$$B + B = 6$$

$$2B = 6$$

$$B = 3$$

$$A = -3$$

$$C = 3$$

Substitute -3 for A , 3 for B and 3 for C in equation (1).

$$\begin{aligned} \frac{6}{(s^2+1)(s+1)} &= \frac{-3s+3}{s^2+1} + \frac{3}{s+1} \\ &= \frac{-3s}{s^2+1} + \frac{3}{s^2+1} + \frac{3}{s+1} \end{aligned}$$

Take inverse Laplace transform to the above equation.

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{6}{(s^2+1)(s+1)} \right] &= \mathcal{L}^{-1} \left(\frac{-3s}{s^2+1} \right) + \mathcal{L}^{-1} \left(\frac{3}{s^2+1} \right) + \mathcal{L}^{-1} \left(\frac{3}{s+1} \right) \\ &= -3 \cos tu(t) + 3 \sin tu(t) + 3e^{-t}u(t) \\ &= (-3 \cos t + 3 \sin t + 3e^{-t})u(t) \end{aligned}$$

Therefore, the Laplace transform of the function $X(s)$ is $\boxed{(-3 \cos t + 3 \sin t + 3e^{-t})u(t)}$.

Step 3 of 6

(b)

The function $X(s)$ is,

$$X(s) = \frac{se^{-\pi s}}{(s^2+1)}$$

Take inverse Laplace transform to the above equation.

Refer table 15.1 in the textbook for time shift property of Laplace transform.

$$\mathcal{L}^{-1} \left[e^{-as} F(s) \right] = f(t-a)u(t-a)$$

$$\mathcal{L}^{-1} \left[\frac{s}{s^2+1} \right] = \cos t$$

$$\mathcal{L}^{-1} \left[\frac{se^{-\pi s}}{(s^2+1)} \right] = \cos(t-\pi)u(t-\pi)$$

Therefore, the Laplace transform of the function $X(s)$ is $\boxed{\cos(t-\pi)u(t-\pi)}$.

Step 4 of 6

(c)

The function $X(s)$ is,

$$X(s) = \frac{8}{s(s+1)^3}$$

Use partial fractions to simplify the function $X(s)$.

$$\frac{8}{s(s+1)^3} = \frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2} + \frac{D}{(s+1)^3} \dots\dots (4)$$

$$8 = A(s+1)^3 + Bs(s+1)^2 + Cs(s+1) + Ds$$

$$8 = (A+B)s^3 + (3A+2B+C)s^2 + (3A+B+C+D)s + A \dots\dots (5)$$

Step 5 of 6

Compare constant terms in equation (5).

$$A = 8$$

Compare coefficient of s^3 terms in equation (5).

$$A + B = 0$$

$$B = -8$$

Compare coefficient of s^2 terms in equation (5).

$$3A + 2B + C = 0$$

$$24 - 16 + C = 0$$

$$C = -8$$

Compare coefficient of s terms in equation (5).

$$3A + B + C + D = 0$$

$$24 - 8 - 8 + D = 0$$

$$D = -8$$

Substitute 8 for A , -8 for B , -8 for C and -8 for D in equation (4).

$$\frac{8}{s(s+1)^3} = \frac{8}{s} - \frac{8}{(s+1)} - \frac{8}{(s+1)^2} - \frac{8}{(s+1)^3} \dots\dots (6)$$

Step 6 of 6

Take inverse Laplace transform to the equation (6).

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{8}{s(s+1)^3}\right] &= \mathcal{L}^{-1}\left(\frac{8}{s}\right) - \mathcal{L}^{-1}\left(\frac{8}{s+1}\right) - \mathcal{L}^{-1}\left(\frac{8}{(s+1)^2}\right) - \mathcal{L}^{-1}\left(\frac{8}{(s+1)^3}\right) \\ &= 8u(t) - 8e^{-t}u(t) - 8te^{-t}u(t) - 4t^2e^{-t}u(t) \\ &= 8u(t)(1 - e^{-t} - te^{-t} - 0.5t^2e^{-t})\end{aligned}$$

Therefore, the Laplace transform of the function $X(s)$ is $\boxed{8u(t)(1 - e^{-t} - te^{-t} - 0.5t^2e^{-t})}$.